

Environmental Studies & IT - Full Answers

1. Sustainable Economic Growth

Introduction:

Sustainable economic growth means development that meets present needs without affecting future generations. It balances economic progress, environmental protection, and social welfare.

Key Features: Long-term economic growth Efficient use of resources Environmental protection Poverty reduction Improved quality of life **Key Sectors:** **Agriculture:** Organic farming, water conservation **Industry:** Eco-friendly manufacturing and recycling **Energy:** Renewable energy like solar and wind **Service Sector:** IT, education, healthcare **Transport:** Public transport and electric vehicles **Conclusion:** Sustainable growth ensures economic progress while protecting resources for future generations.

2. AI-Driven Solutions for Water Management

Introduction:

AI improves water management through smart monitoring, prediction, and automation.

Applications of AI: **Smart Monitoring:** Tracks water quality and usage **Leak Detection:** Identifies pipeline leaks quickly **Flood Prediction:** Provides early warning systems **Smart Irrigation:** Saves water in agriculture **Demand Prediction:** Predicts future water needs **Advantages:** Reduces water wastage Improves efficiency Supports sustainability Cost savings **Conclusion:** AI helps conserve water and improve sustainable water management systems.

3. E-Waste: Types, Sources, Environmental and Health Impacts

Introduction:

E-waste refers to discarded electronic devices such as computers, mobiles, and TVs.

Types of E-Waste: Household appliances IT and communication devices Consumer electronics Medical equipment Lighting equipment **Sources:** Households Industries Offices Hospitals Educational institutions **Environmental Impacts:** Soil pollution Water pollution Air pollution Loss of natural resources **Health Impacts:** Respiratory diseases Skin irritation Nervous system damage Cancer risks **Conclusion:** Proper recycling and disposal are essential to reduce the harmful effects of e-waste.

4. Circular Economy and Role of IT

Introduction:

A circular economy focuses on reducing waste through reuse, recycling, and recovery of resources.

Principles: Reduce Reuse Recycle Recover **Role of IT:** Smart waste management systems IoT-enabled monitoring AI for waste sorting Big data analytics Digital sharing platforms Blockchain for supply chain transparency **Conclusion:** IT supports sustainable resource management and promotes circular economy practices.

5. Recycling and Safe Disposal of IT Equipment

Introduction:

Improper disposal of IT equipment causes pollution and health hazards.

Recycling Techniques: Collection and segregation Reuse and refurbishment Material recovery Mechanical recycling Chemical recycling Plastic recycling **Safe Disposal Methods:** Authorized recycling centers Secure data destruction Safe battery disposal Avoid landfill dumping Producer take-back programs **Benefits:** Conserves resources Reduces pollution Saves energy Supports sustainability **Conclusion:** Proper recycling and disposal methods help protect the environment and human health.

6. Green Buildings

Introduction:

Green buildings are environmentally responsible structures that use resources efficiently.

Features: Energy efficiency Water conservation Eco-friendly materials Natural lighting and ventilation Waste management systems **Principles:** Reduce resource consumption Reuse and recycle materials Protect human health Minimize environmental impact Improve efficiency **Criteria:** Energy performance Water efficiency Indoor environmental quality Sustainable material use Waste reduction **Conclusion:** Any building can become a green building through sustainable design and efficient resource management.